Linear Algebra & Calculus 2420030390

CO-1 Home Assignment

1. Calculate the rank of the matrix by reducing into echelon form

(i) $A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 4 & 8 & 2 & 0 \\ 4 & -2 & -3 & -1 & 0 \\ 9 & 6 & 7 & 2 & 0 \\ 6 & 8 & 8 & 6 & 0 \end{bmatrix}$

Sol: (i) $A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$

$R_1 \Leftrightarrow R_2 \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix}$

$R_3 \to R_3 - 3R_1 \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{bmatrix}$

$R_3 \to R_3 + 5R_2 \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix}$

Rank of A is 3

(ii) $A = \begin{bmatrix} 1 & 4 & 8 & 2 & 0 \\ 4 & -2 & -3 & -1 & 0 \\ 9 & 6 & 7 & 2 & 0 \\ 6 & 8 & 8 & 6 & 0 \end{bmatrix}$

$R_2 \to R_2 - 4R_1$
$R_3 \to R_3 - 9R_1$
$R_4 \to R_4 - 6R_1$
$\begin{bmatrix} 1 & 4 & 3 & 2 & 0 \\ 0 & -18 & -15 & -9 & 0 \\ 0 & -30 & -20 & -16 & 0 \\ 0 & -18 & -15 & 12 & 0 \end{bmatrix} \simeq \begin{bmatrix} 1 & 4 & 3 & 2 \\ 0 & -6 & -5 & -3 \\ 0 & -15 & -10 & -8 \\ 0 & -9 & -5 & 6 \end{bmatrix}$

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$R_3 \to (R_3)2-(R_2)5$
$R_4 \to (R_4)2-(R_2)3$
$$\begin{bmatrix} 1 & 4 & 3 & 2 & 0 \\ 0 & -6 & -5 & -3 & 0 \\ 0 & 0 & 5 & -1 & 0 \\ 0 & 0 & 5 & 17 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 3 & 2 & 0 \\ 0 & -6 & -5 & -3 & 0 \\ 0 & 0 & 5 & -1 & 0 \\ 0 & 0 & 5 & 7 & 0 \end{bmatrix}$$

$R_4 \to R_4 - R_3$
$$\begin{bmatrix} 1 & 4 & 3 & 2 & 0 \\ 0 & -6 & -5 & -3 & 0 \\ 0 & 0 & 5 & -1 & 0 \\ 0 & 0 & 0 & 18 & 0 \end{bmatrix}$$

$\therefore$ Rank is 4

2. In a cricket match, chennai super kings needed just 6 runs to win with 1 ball left to go in the last one the last ball was bowled & the batsman at the grease hit it high up the ball tranversed along a path in a vertical plane & the eqn of the path is $y = px^2 + qx + r$ with respect to the path is in the vertical plane xy-coordinate system & the ball traversed through the points (10, 8), (20, 16), (30, 18) can you conclude that chennai super kings won the match? use gauss elimination method?

Sol: Given eq $y = px^2 + qx + r$

for (10, 8) $8 = p(10)^2 + q(10) + r$

$100p + 10q + r = 8$

(20, 16) $16 = p(20)^2 + q(20) + r$

$400p + 20q + r = 16$

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$(30, 18)$ $\quad 18 = p(30)^2 + q(30) + r$

$$900p + 30q + r = 18$$

$AX = B$

$$\begin{bmatrix} 100 & 10 & 1 \\ 400 & 20 & 1 \\ 900 & 30 & 1 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 8 \\ 16 \\ 18 \end{bmatrix}$$

$$A|B = \begin{bmatrix} 100 & 10 & 1 & 8 \\ 400 & 20 & 1 & 16 \\ 900 & 30 & 1 & 18 \end{bmatrix}$$

$$A|B = \begin{bmatrix} 100 & 10 & 1 & 8 \\ 400 & 20 & 1 & 16 \\ 900 & 30 & 1 & 8 \end{bmatrix}$$

$R_2 \to R_2 - 4R_1$
$R_3 \to R_3 - 9R_1$

$$\begin{bmatrix} 100 & 10 & 1 & 8 \\ 0 & -20 & -3 & -16 \\ 0 & -60 & -8 & -54 \end{bmatrix}$$

$R_3 \to R_3 - 3R_2$

$$\begin{bmatrix} 100 & 10 & 1 & 8 \\ 0 & -20 & -3 & -16 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

Rank of $A|B = 3 =$ Rank of A

It is consistant & has unique solution

$$\begin{bmatrix} 100 & 10 & 1 \\ 0 & -20 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 8 \\ -16 \\ -6 \end{bmatrix}$$

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$(r)(1) = -6$ $\qquad -20q - 3r = -16$

$r = -6$ $\qquad -20q + 18 = -16$

$-20q = -16 - 18 = -34$

$q = 34/20$

$q = \frac{17}{10}$

$100p + 10q + r = 8$

$100p + 10\left[\frac{17}{10}\right] - 6 = 8$

$100p + 17 - 6 = 8$

$100P = 8 - 11 = 3$

$P = \frac{-3}{100}$

$y = px^2 + qx - r$

$y = \left[\frac{-3}{100}\right]x^2 + \left[\frac{17}{10}\right]x - 6$

In order to conclude that chennai super kings won the match, we need to check if the final position of the ball is sufficient for 6 runs.

Since vertical condinate of ball suggests it was hit high (six) & if we assume the boundary criteria are met, we can conclude that chennai super kings won the match this shot

3. A youth group is selling snacks to raise money to attend their convention Any sold 2 pounds of cavely, 3 boxes of cookies & 1 can of poplorn for a total sale of \$65 Brian sold 4 pounds of candy, 6 boxes of

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cookies & 3 cans of popcorn for a total sale of \$140. panlia sold 8 pounds of candy, 8 boxes of cookies & 5 cans of popcorn for a total sale of \$250 Determine the cost of each item using Gauss elimination method.

sol: let x, y, z = be the candy, cookies & popcorn respectively

		x	y	z	Total
A	x	2	3	1	65
B	y	4	6	3	140
P	z	8	8	5	250

$$\begin{bmatrix} 2 & 3 & 1 \\ 4 & 6 & 3 \\ 8 & 8 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 65 \\ 140 \\ 250 \end{bmatrix}$$

$AX = B$

$$A|B = \begin{bmatrix} 2 & 3 & 1 & 65 \\ 4 & 6 & 3 & 140 \\ 8 & 8 & 5 & 250 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - 4R_1$

$$\begin{bmatrix} 2 & 3 & 1 & 65 \\ 0 & 0 & 1 & 10 \\ 0 & -4 & 1 & -10 \end{bmatrix}$$

$R_2 \Leftrightarrow R_3$

$$\begin{bmatrix} 2 & 3 & 1 & 65 \\ 0 & -4 & 1 & -10 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

Rank of $A|B = 3 =$ Rank of A 2420030390

It is consistant & unique solution.

$$\begin{bmatrix} 2 & 3 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 65 \\ -10 \\ 10 \end{bmatrix}$$

$2x+3y+z=65$

$2x+15+10=65$

$2x=65-25$

$2x=40$

$x=20$

$(1)(z)=10$

$z=10$

$-4y+z=-10$

$-4y+10=-10$

$-4y=-20$

$y=5$

$\therefore x=20, y=5, z=10$

solve the following by using LU- decomposition method

(i) $x+y+z=1$, $3x+y-3z=5$, $x-2y-5z=10$

(ii) $x+y-z=4$, $x-2y+3z=-6$, $2x+3y+z=7$

$\therefore$ (i) $x+y+z=1$

$3x+y-3z=5$

$x-2y-5z=10$

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -3 \\ 1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 10 \end{bmatrix}$$

$AX=B$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -3 \\ 1 & -2 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

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$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23} \\ l_{31}u_{11} & l_{31}u_{12}+u_{22}l_{32} & l_{31}u_{13}+l_{32}u_{23}+u_{33} \end{bmatrix}$$

$\boxed{u_{11}=1}$ $\boxed{u_{12}=1}$ $\boxed{u_{13}=1}$

$l_{21}u_{11}=3$
$l_{21}=3/1$
$\boxed{l_{21}=3}$

$l_{21}u_{12}+u_{22}=1$
$(3)(1)+u_{22}=1$
$u_{22}=1-3$
$\boxed{u_{22}=-2}$

$l_{21}u_{13}+u_{23}=-3$
$(3)(1)+u_{23}=-3$
$u_{23}=(-3)-3$
$\boxed{u_{23}=-6}$

$l_{31}u_{11}=1$
$l_{31}=1/1$
$\boxed{l_{31}=1}$

$l_{31}u_{12}+u_{22}l_{32}=2$
$(1)(1)+(-2)l_{32}=-2$
$1+(-2)l_{32}=-2$
$l_{32}=(-2-1)/-2$
$l_{32}=-3/-2$
$l_{32}=3/2$

$l_{31}u_{13}+l_{32}u_{23}+u_{33}=-5$
$(1)(1)+\left[\frac{3}{\cancel{2}}\right]\left[\cancel{-6}^{\,3}\right]+u_{33}=-5$
$1+(-9)+u_{33}=-5$
$u_{33}=-5+9-1=9-6=3$
$\boxed{u_{33}=3}$

$LUX=B$
$UX=Y$ where $Y=\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -6 \\ 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$\therefore LY=B$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 3/2 & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}=\begin{bmatrix} 1 \\ 5 \\ 10 \end{bmatrix}$$

$y_1=1$

$3y_1+y_2=5$
$y_2=5-3$
$y_2=2$

$y_1+\frac{3y_2}{2}+y_3=10$
$1+3\cdot\frac{\cancel{2}}{\cancel{2}}+y_3=10$
$y_3=10-4=6$

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$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}$$

$x+y+z=1$ $\quad$ $-2y-6z=2$ $\quad$ $3z=6$

$x+(-7)+2=1$ $\quad$ $-2y-6(2)=2$ $\quad$ $z=2$

$x=1+7-2$ $\quad$ $-2y=2+12$

$x=6$ $\quad$ $-2y=14$

$y=-7$

$$\therefore x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -7 \\ 2 \end{bmatrix}$$

(ii) $x+y-z=A$

$x-2y+3z=-6$

$2x+3y+z=7$

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ 7 \end{bmatrix}$$

$Ax=B$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23} \\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33} \end{bmatrix}$$

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$\boxed{U_{11}=1}$ $\boxed{U_{12}=1}$ $\boxed{U_{13}=-1}$

$L_{21}U_{11}=1$

$L_{21}=1/1$

$\boxed{L_{21}=1}$

$L_{21}U_{12}+U_{22}=-2$

$(1)(1)+U_{22}=-2$

$\boxed{U_{22}=-3}$

$L_{21}U_{13}+U_{23}=3$

$(1)(-1)+U_{23}=3$

$\boxed{U_{23}=4}$

$L_{31}U_{11}=2$

$L_{31}=2/1$

$\boxed{L_{31}=2}$

$L_{31}U_{12}+L_{32}U_{22}=3$

$(2)(1)+L_{32}(-3)=3$

$L_{32}(-3)=3-2=1$

$\boxed{L_{32}=-1/3}$

$L_{31}U_{13}+L_{32}U_{23}+U_{33}=$

$(2)(-1)+\left(\frac{-1}{3}\right)(4)U_3$

$-6-4+3U_{33}=3$

$3U_{33}=3+10$

$\boxed{U_{33}=13/3}$

$$\begin{bmatrix}1&0&0\\1&1&0\\2&-1/3&1\end{bmatrix}\begin{bmatrix}1&1&-1\\0&-3&4\\0&0&13/3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}4\\-6\\-7\end{bmatrix}$$

$LUX=B$

$UX=Y$ where $Y=\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$

$$\begin{bmatrix}1&0&0\\1&1&0\\2&-1/3&1\end{bmatrix}\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}=\begin{bmatrix}4\\-6\\7\end{bmatrix}$$

$y_1=4$

$y_1+y_2=-6$

$y_2=-6-4$

$y_2=-10$

$2y_1-\frac{y_2}{3}+y_3=7$

$2(4)-\left(\frac{-10}{3}\right)+y_3=7$

$(8\cdot 3)+10+3y_3$

$3y_3=21-10-24$

$\boxed{y_3=-13/3}$

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$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ -13/3 \end{bmatrix}$$

$\frac{13}{3} z = \frac{-13}{3}$

$z = -1$

$-3y + 4z = -10$

$-3y + 4(-1) = -10$

$-3y = -10 + 4$

$-3y = -6$

$y = 2$

$x + y + z = 4$

$x + 2 - (-1) = 4$

$x = 1$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Determine the eigen values & eigen vector of matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

characteristic eqn $(A - \lambda I) = \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix}$

By eq. $\lambda^3 - \lambda^2(\text{trace of } A) + \lambda(\text{sum of minors of principle elements}) - \det A = 0$

$\lambda^3 - \lambda^2(6+3+3) + \lambda[(9-1) + (18-4) + (18-4)]$
$- [6(9-1) + 2(-6+2) + 2(2-6)] = 0$

$\lambda^3 - 12\lambda^2 + \lambda[8 + 14 + 4] - 1[6(8) + 2(-4) + 2(-4)] = 0$

$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

$\lambda = 8, 2, 2$

$\therefore$ eigen values $8, 2, 2$

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case(ii): Eigen vector corresponding to Eigen value $\lambda = 8$

$$(A-8I) = \begin{bmatrix} 6-8 & -2 & 2 \\ -2 & 3-8 & -1 \\ 2 & -1 & 3-8 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}$$

$R_2 \to R_2 - R_1$
$R_3 \to R_3 + R_1$

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}$$

$R_3 \to R_3 + R_2$

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\rho(A-8I) = 2 <$ no of unknowns

choose $n-r = 3-2 = 1$ variable independent

$$(A-\lambda I)(X) = 0$$

$$\begin{bmatrix} -2 & -2 & 2 \\ 0 & -3 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $y = k$

$-3y - z = 0$
$-3y = z$
$z = -3k$

$-2x - 2y + 2z = 0$
$-2x - 2k + 2(-3k) = 0$
$-2x - 2k - 6k = 0$
$-2x - 8k = 0$
$-2x = 8k$
$x = -4k$

$$X_1 = \begin{bmatrix} -4k \\ k \\ -3k \end{bmatrix} = k \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix}$$

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∴ Eigen vectors are:

$$x_1 = \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad x_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

6. verify the system $dx/dt = Ax$ is stable or not where $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

sol: characteristic eqn $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix}$$

By eqn $\lambda^3 - \lambda^2(\text{trace of } A) + \lambda(\text{sum of minor principle elements}) - \det A = 0$

$$\lambda^3 - \lambda^2(1+2+3) + \lambda[(6-2)+(3+2)+(2)]$$

$$-[(1)(6-2) + 0 - 1(2-4)] = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda = 1, 3, 2$$

∴ since all the eigen values are positive

∴ the system is not stable

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chek whether matrix A is diagonalize or not where $A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 1 & 4 & -3 \end{bmatrix}$

ok: characteristic ean $|A-\lambda I|=0$

$$(A-\lambda I) = \begin{bmatrix} 5-\lambda & 0 & 0 \\ 0 & 5-\lambda & 0 \\ 1 & 4 & -3-\lambda \end{bmatrix}$$

By ean $\lambda^3 - \lambda^2$ (trace of A) $+ \lambda$ (sum of minor principle elements) $-$ det A $= 0$

$$\lambda^3 - \lambda^2(5+5-3) + \lambda((-15)+(-15)+25] - [5(-15)] = 0$$

Algebraic multiplicit of $5=2$

$$\lambda^3 - 7\lambda^2 - 5\lambda + 75 = 0$$

$$\lambda = -3, 5, 5$$

case(i): Eigen vector corresponding to Eigen value $\lambda = -3$

$$(A-(-3)I) = (A+3I) = \begin{bmatrix} 5+3 & 0 & 0 \\ 0 & 5+3 & 0 \\ 1 & 4 & -3+3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 1 & 4 & 0 \end{bmatrix}$$

$R_2 \leftrightarrow R_3$ $\begin{bmatrix} 8 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 8 & 0 \end{bmatrix}$

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$R_2 \to 8(R_2) - R_1$ $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\rho(A+3I) = 2 <$ no of unknowns

choose $n-r = 3-2 = 1$ variable independent

$$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

let $z = k$ $\quad 8x = 0 \quad 4y = 0$

$x = 0 \quad y = 0$

$$X_1 = \begin{bmatrix} 0 \\ 0 \\ k \end{bmatrix} = k \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

case(ii): Eigen vector corresponding to Eigen value $\lambda = 5$

$$(A-5I) = \begin{bmatrix} 5-5 & 0 & 0 \\ 0 & 5-5 & 0 \\ 1 & 4 & 3-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 4 & -8 \end{bmatrix}$$

$\rho(A-5I) = 1 <$ no of unknows

choose $n-r = 3-1 = 2$ variable independent

$$(1)(x) + 4(y) + (-8)(z) = 0$$

$$x + 4k_1 - 8k_2 = 0$$

$$x = 8k_2 - 4k_1$$

$$X = \begin{bmatrix} 8k_2 - 4k_1 \\ k_1 \\ k_2 \end{bmatrix} = k_1 \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix} + k_2 \begin{bmatrix} 8 \\ 0 \\ 1 \end{bmatrix}$$

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X_1 & X_2 are orthogonal.

Geometric multiplicity 2

∴ Since Algebraic multiplicity = Geometric multiplicity

the matrix is diagonitable

Determine the Eigen values of matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 0 \\ 1 & 2 & -4 \end{bmatrix}$$ & hence determine A^3, A^T, A^{-1} also verify the Eigen values of sum & product of Eigen values is same as trace & determinant of matrix A

To find Eigen value characteristic eqn

characteristic matrix $(A-\lambda I)=0$

$$(A-\lambda I) = \begin{bmatrix} 1-\lambda & 0 & 0 \\ -2 & 3-\lambda & 0 \\ 1 & 2 & -4-\lambda \end{bmatrix}$$

$$(A-\lambda I) = \begin{bmatrix} 1-\lambda & 0 & 0 \\ -2 & 3-\lambda & 0 \\ 1 & 2 & -4-\lambda \end{bmatrix}$$

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By eqn $\lambda^3 - \lambda^2(\text{trace of } A) + \lambda[\text{sum of minors of}$ principle diagonal elements$] - \det A = 0$

$$\lambda^3 - \lambda^2(1+3-4) + \lambda[(-12)+(-4)+3] - [1(-12)$$

$$\lambda^3 + \lambda[-16+3] - (-12) = 0$$

$$\lambda^3 - 13\lambda + 12 = 0$$

$$\lambda = -4, 3, 1$$

$\therefore$ Eigen values are $-4, 3, 1$

$\therefore$ Eigen values of A^3 are $\lambda_1^3, \lambda_2^3, \lambda_3^3$ which is equal to $(-4)^3, (3)^3, (1)^3 = -64, 27, 1$

$\therefore$ Eigen values of A^T are same as that of A $-4, 3, 1$

$\therefore$ Eigen values of A^{-1} are $1/\lambda_1, 1/\lambda_2, 1/\lambda_3$ which is equal to $-1/4, 1/3, 1$

Sum of Eigen values $= -4+3+1 = 0$ trace of A

product of Eigen values $= (-4)(3)(1) = -12 = \det A$

$\therefore$ sum & product of Eigen values is same as trace & determinant of matrix A

9. Reduce the given quadratic form $Q = x_1^2 + 5x_2^2 + x_3^2 + 2x_2x_3 + 6x_1x_3$

Sol: Given eq $x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_1x_3$

	x_1	x_2	x_3
x_1	1	1	3
x_2	1	5	1
x_3	3	1	1

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

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To find eigen values & vectors, characteristic eqn $(A-\lambda I)=0$, characteristic matrix $(A-\lambda I)=0$

$$(A-\lambda I)=\begin{bmatrix}1-\lambda & 1 & 3\\ 1 & 5-\lambda & 1\\ 3 & 1 & 1-\lambda\end{bmatrix}$$

$$(A-\lambda I)=\begin{bmatrix}1-\lambda & 1 & 3\\ 1 & 5-\lambda & 1\\ 3 & 1 & 1-\lambda\end{bmatrix}$$

By eqn $\lambda^3-\lambda^2(\text{trace of } A)+\lambda(\text{sum of minors of principle diagonal elements})-\det A=0$

$$\lambda^3-\lambda^2(1+5+1)+\lambda[(5-1)+(1-9)+(5-1)]$$
$$-[(1)(5-1)-1(1-3)+3(1-15)]=0$$

$$\lambda^3-7\lambda^2+36=0$$
$$\lambda=-2,6,3$$

case(i): Eigen vector corresponding to eigen value $\lambda=-2$

$$(A-(-2)I)=(A+2I)=\begin{bmatrix}1+2 & 1 & 3\\ 1 & 5+2 & 1\\ 3 & 1 & 1+2\end{bmatrix}$$

$$=\begin{bmatrix}3 & 1 & 3\\ 1 & 7 & 1\\ 3 & 1 & 3\end{bmatrix}$$

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$R_2 \to (R_2)3 - R_1$
$R_3 \to R_3 - R_1$
$$\begin{bmatrix} 3 & 1 & 3 \\ 0 & 20 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\rho(A+2I) = 2 <$ no of unknowns

choose $n-r = 3-2 = 1$ variable independent

$(A+2I)(X) = 0$

$$\begin{bmatrix} 3 & 1 & 3 \\ 0 & 20 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$3x + y + 3z = 0$

let $z = k$ $\quad 20y = 0$ $\quad 3x + 3k = 0$

$y = 0$ $\quad 3x = -3k$

$x = -k$

$$X_1 = \begin{bmatrix} -k \\ 0 \\ k \end{bmatrix} = k \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

case(ii): Eigen vector corresponding to Eigen value $\lambda = 6$

$$(A-6I) = \begin{bmatrix} 1-6 & 1 & 3 \\ 1 & 5-6 & 1 \\ 3 & 1 & 1-6 \end{bmatrix} = \begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

$R_2 \to 5R_2 + R_1$
$R_3 \to 5R_3 + 3R_1$
$$\begin{bmatrix} -5 & 1 & 3 \\ 0 & -4 & 8 \\ 0 & 8 & -16 \end{bmatrix}$$

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$$\begin{bmatrix} -5 & 1 & 3 \\ 0 & -4 & 8 \\ 0 & 8 & -16 \end{bmatrix}$$

$R_2 \to R_3 + 2R_2$ $\begin{bmatrix} -5 & 1 & 3 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix}$

$\rho(A-6I) = 2 <$ no of unknowns

choose $n-r = 3-2 = 1$ variable independent

$$\begin{bmatrix} -5 & 1 & 3 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

let $z = k$ $\quad -4y + 8z = 0$ $\qquad -5x + y + 3z = 0$

$-4y = -8z$ $\qquad -5x + 2k + 3k = 0$

$y = 2z$ $\qquad -5x + 5k = 0$

$y = 2k$ $\qquad x = k$

$$x_2 = \begin{bmatrix} k \\ 2k \\ k \end{bmatrix} = k\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

case (iii): Eigen vector corresponding to eigen value $\lambda = 3$

$$(A-3I) = \begin{bmatrix} 1-3 & 1 & 3 \\ 1 & 5-3 & 1 \\ 3 & 1 & 1-3 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix}$$

$R_2 \to 2R_2 + R_1$

$R_3 \to 2R_3 + 3R_1$ $\begin{bmatrix} -2 & 1 & 3 \\ 0 & 5 & 5 \\ 0 & 5 & 5 \end{bmatrix}$

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$R_3 \to R_3 - R_2 \begin{bmatrix} -2 & 1 & 3 \\ 0 & 5 & 5 \\ 0 & 0 & 0 \end{bmatrix}$

$\rho(A-3I) = 2 <$ no of unknowns

choose $n-r = 3-2 = 1$ variable independently

$(A-3I)(x) = 0$

$$\begin{bmatrix} -2 & 1 & 3 \\ 0 & 5 & 5 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

let $z = k$ $\quad 5y + 5z = 0$ $\quad -2x + y + 3z = 0$

$y = z$ $\quad -2x - k + 3k = 0$

$y = -k$ $\quad -2x + 2k = 0$

$x = k$

$$x_3 = \begin{bmatrix} k \\ -k \\ k \end{bmatrix} = k\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$\therefore x_1, x_2, x_3$ are pair-wise orthogonal

$\|x_1\| = \sqrt{(-1)^2 + (0)^2 + (1)^2} = \sqrt{2}$

$\|x_2\| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$

$\|x_3\| = \sqrt{(1)^2 + (-1)^2 + (1)^2} = \sqrt{3}$

$$R_1 = \frac{x_1}{\|x_1\|} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

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$$e_2 = \frac{x_2}{\|x_2\|} = \begin{pmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix}$$

$$e_3 = \frac{x_3}{\|x_3\|} = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$P = [e_1 \; e_2 \; e_3]$$

$$P = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

$$P^T = \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$P^TAP = D = \begin{bmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{6} & \\ 0 & 2/\sqrt{6} & \\ 1/\sqrt{2} & 1/\sqrt{6} & \end{bmatrix}$$

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

canonical form $= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$

$= (-2)y_1^2 + (6)y_2^2 + (3)y_3^2$

$= -2y_1^2 + 6y_2^2 + 3y_3^2$

$\therefore$ Rank is 3

index is 2

signature 1

nature is indefinite

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10. Reduce the Q.F $x^2+y^2+z^2+2xy+yz-6zx$ into sum of squares Also specify the nature of the Q.F

$x^2+y^2+z^2+2xy+4yz-6zx$

sol: Given

	x	y	z
x	1	1	-3
y	1	1	2
z	-3	2	1

$$A=\begin{bmatrix} 1 & 1 & -3 \\ 1 & 1 & 2 \\ -3 & 2 & 1 \end{bmatrix}$$

To find eigen values & vectors, characteristic matrix is ean $(A-\lambda I)=0$, characteristic

$(A-\lambda I)=0$

$$(A-\lambda I)=\begin{bmatrix} 1-\lambda & 1 & -3 \\ 1 & 1-\lambda & 2 \\ -3 & 2 & 1-\lambda \end{bmatrix}$$

$$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 1 & -3 \\ 1 & 1-\lambda & 2 \\ -3 & 2 & 1-\lambda \end{vmatrix}$$

By ean $\lambda^3-\lambda^2(\text{trace of } A)+\lambda(\text{sum of minors of principle elements})-\det A=0$

$$\lambda^3-\lambda^2(1+1+1)+\lambda[(1-4)+(1-9)+(1-1)]$$
$$-[1(1-4)-1(1+6)-3(2+3)]=0$$

$$\lambda^3-3\lambda^2-11\lambda+25=0$$

3/2/25